

Implement and demonstrate the FIND-S algorithm for finding the most specific hypothesis based on a given set of training data samples.

Aim

To implement and demonstrate the **FIND-S algorithm** for finding the **most specific hypothesis** from a given set of training data samples.

Apparatus Required

- Google Colab

Theory

The **FIND-S algorithm** is a supervised machine learning algorithm used to determine the **most specific hypothesis** that fits all positive training examples.

- It considers **only positive examples**.
- It ignores negative examples.
- It starts with the most specific hypothesis and gradually generalizes it whenever necessary.

Algorithm

1. Read the training dataset.
2. Initialize the hypothesis with the most specific values (0).
3. Consider only positive training examples.
4. For each positive example:
 - Replace 0 with the corresponding attribute value.
 - If attribute values differ, replace with ?.
5. Ignore negative examples.
6. Display the final hypothesis.

Code :

```
import pandas as pd
```

```
data = pd.DataFrame({  
    'Sky': ['Sunny', 'Sunny', 'Rainy', 'Sunny'],  
    'Temp': ['Warm', 'Warm', 'Cold', 'Warm'],  
    'Humidity': ['Normal', 'High', 'High', 'High'],  
    'Wind': ['Strong', 'Strong', 'Strong', 'Strong'],  
    'Play': ['Yes', 'Yes', 'No', 'Yes']  
})
```

```
X = data.iloc[:, :-1].values
```

```
Y = data.iloc[:, -1].values
```

```
h = ['0'] * len(X[0])
```

```
for i in range(len(Y)):
```

```
    if Y[i] == 'Yes':
```

```
        for j in range(len(h)):
```

```
            if h[j] == '0':
```

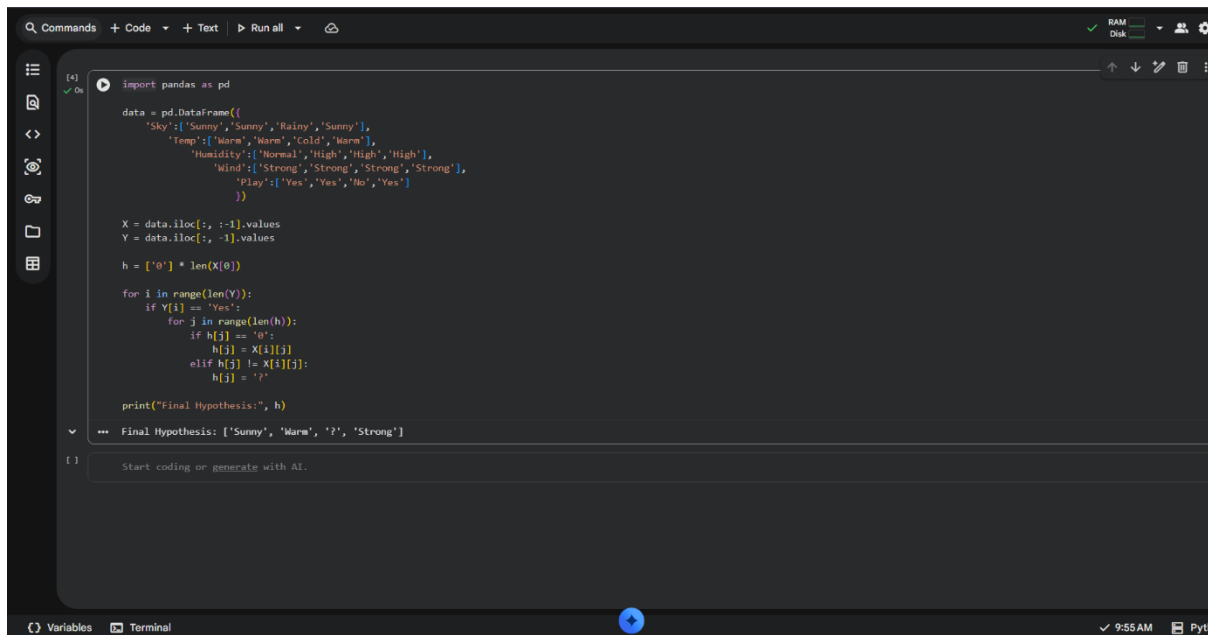
```
                h[j] = X[i][j]
```

```
            elif h[j] != X[i][j]:
```

```
                h[j] = '?'
```

```
print("Final Hypothesis:", h)
```

Output:



```
[4] In: import pandas as pd

data = pd.DataFrame({
    'Sky': ['Sunny', 'Sunny', 'Rainy', 'Sunny'],
    'Temp': ['Warm', 'Warm', 'Cold', 'Warm'],
    'Humidity': ['Normal', 'High', 'High', 'High'],
    'Wind': ['Strong', 'Strong', 'Strong', 'Strong'],
    'Play': ['Yes', 'Yes', 'No', 'Yes']
})

X = data.iloc[:, :-1].values
Y = data.iloc[:, -1].values

h = ['0'] * len(X[0])

for i in range(len(Y)):
    if Y[i] == 'Yes':
        for j in range(len(h)):
            if h[j] == '0':
                h[j] = X[i][j]
            elif h[j] != X[i][j]:
                h[j] = '?'

print("Final Hypothesis:", h)

Out: Final Hypothesis: ['Sunny', 'Warm', '?', 'Strong']

[5] In: Start coding or generate with AI.
```

Result

The FIND-S algorithm was successfully implemented. The algorithm identified the most specific hypothesis that is consistent with all the positive training examples.